XRPL Validator — Daily Engineering Report · 2026-11-02

Period: 2026-11-02 00:00 to 24:00 US Eastern (2026-11-02 05:00 to 2026-11-03 05:00 UTC); ledgers #108,060,072 to #108,081,805. Every count from the ledger and the validator's logs stops at #108,081,805, closed 04:59:52 UTC. Generated 2026-11-03 00:20 EST by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Status

Normal — coverage outside planned maintenance 100.000% (objective 99.9%), every validation agreed with the network, no open engine receipt.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 21,734 of the 21,734 ledgers the network closed (100.000%); every validation matched the network's validated ledger.
- **Native engine:** 0 engine receipts.
- **Changes:** 0 deployments, 0 commits merged into main.
- **Network:** 2 slow-close periods; the longest, 18:09 UTC–18:09 UTC, closed ledgers up to 18 s apart.
- **Independence program:** 5 of 30 qualifying days (snapshot at 05:21:01 UTC).
- **Amendment fixCleanup3_4_0** holds majority; activation due ~2026-11-04 14:46 UTC.

Service levels

Measure	Value
Ledgers closed by the network	21,734
Validations issued	21,734
Coverage (validations / ledgers closed)	100.000%
Validations matching the network's validated ledger hash	21,734 of 21,734
Unsigned during planned restarts	0
Unsigned outside maintenance	0
Coverage outside planned maintenance (objective 99.9%)	100.000%
Restarts	0
Basis: our own ledger hash	21,734
External agreement, data.xrpl.org (UTC day 2026-11-02)	1.00000 (0 missed of 19,038)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#108,060,072–#108,081,805), by step:

Step	Ledgers with a receipt
Step 1 · native engine	0
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters, snapshot at 05:21:01 UTC (since the last restart, 2026-10-28 22:52 UTC; not cut off at the period's last ledger):

Step	Checked	Result
1 · native engine vs libxrpl	113,810 ledgers, 21,388,826 transactions	113,809 fully matched; results 2, bytes 1, keys 0/0
2 · transaction records	114,402 ledgers, 21,769,553 records	21,769,550 exact; differ 3, tree 3
3 · transaction order	114,441 ledgers (mode: on)	114,439 exact; order 0, results 2, bytes 0, refused 0
4 · ledger data from peers	114,489 ledgers fetched	0 matched; differ 0, failed 0
4 · trusted validations	113,658 ledgers judged	113,651 agreed; differ 0, short of quorum 7
5 · our own ledger hash	114,489 ledgers	114,397 own; differ 3, partial 89
Writer	every ledger written	mismatch 1, distrusted 0

"Partial" ledgers are those the engine could not yet build alone while warming up after a restart.

Receipt register

No receipts this day.

Incidents

No incidents.

Change log

Deployments

No deployments.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Independence program

Qualifying days: **5 of 30** (snapshot at 05:21:01 UTC). A UTC day qualifies when every validation is of a ledger our own engine built, with our own transaction order and ledger data taken from peers, and coverage is at least 98%.

UTC day 2026-11-02 — tracked #108,055,555–#108,077,355	Value
Validations	21,801 (21,801 own, 0 borrowed)
Coverage of the tracked ledgers	100.00%
Our own transaction order	on
Ledger data from peers	yes

Meets every criterion.

The validator saves this record every 64 validations, so a day in progress lags the signing log slightly; it is not cut off at the period's last ledger.

Network

Measure	Value
Ledgers closed	21,734 (#108,060,072–#108,081,805)
Mean close interval	3.98 s
Longest close interval	18 s (before #108,072,029)
Closes at least 15 s apart	2
Empty ledgers	0

- Slow closes 18:09:02 UTC–18:09:20 UTC: ledgers #108,072,029, 1 close at least 15 s apart, up to 18 s.
- Slow closes 02:12:03 UTC–02:12:20 UTC: ledgers #108,079,358, 1 close at least 15 s apart, up to 17 s.

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_4_0	2026-10-21 14:46 UTC	2026-11-04 14:46 UTC

Amendments enabled on mainnet: 97.

Open items

- Amendment fixCleanup3_4_0: activation due ~2026-11-04 14:46 UTC.

Corrections to earlier reports

None.